

Training Free Zero-Shot Image Anomaly Localisation via Diffusion Inversion

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Introduction

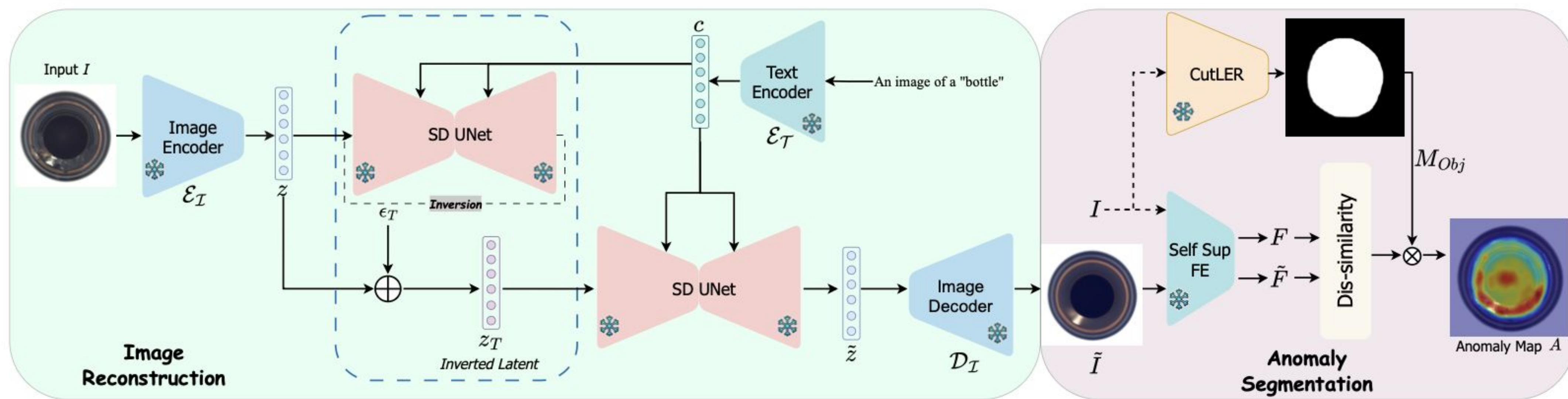
Problem

- Zero-Shot image Anomaly Detection (ZSIAD) aims to detect and localise anomalies **without access to any normal target samples**.
- Current ZSIAD approaches leverage;
 - external data**, e.g., train on MVTec-AD [1] and test on VisA [2],
 - additional modalities such as language to generate **fine-grained prompts**.
- Hence, they require **i) labelled datasets**, and **ii) domain expertise** to curate text prompts.
- Training-free vision only methods largely remain limited to image-level classification.

Contributions

- We introduce a training-free and vision-only ZSIAD framework which relaxes the need for curated text prompts utilizing inversion of a pretrained Denoising Diffusion Implicit Model (DDIM) [3].**
- We conduct experiments on common industrial ZSIAD datasets and show that our approach achieves SoTA performance on VisA.**

Our Method



Inversion Mechanisms

- Inversion of a generative model aim to **find the latents responsible for the generation of a given image**.
- They are frequently used for **image editing**.
- Our intuition** is that as the baseline model is **trained using only normal data**, if we regenerate the image using inverted latents, it should be free from any **anomalies**.

Our overall workflow comprises four stages:

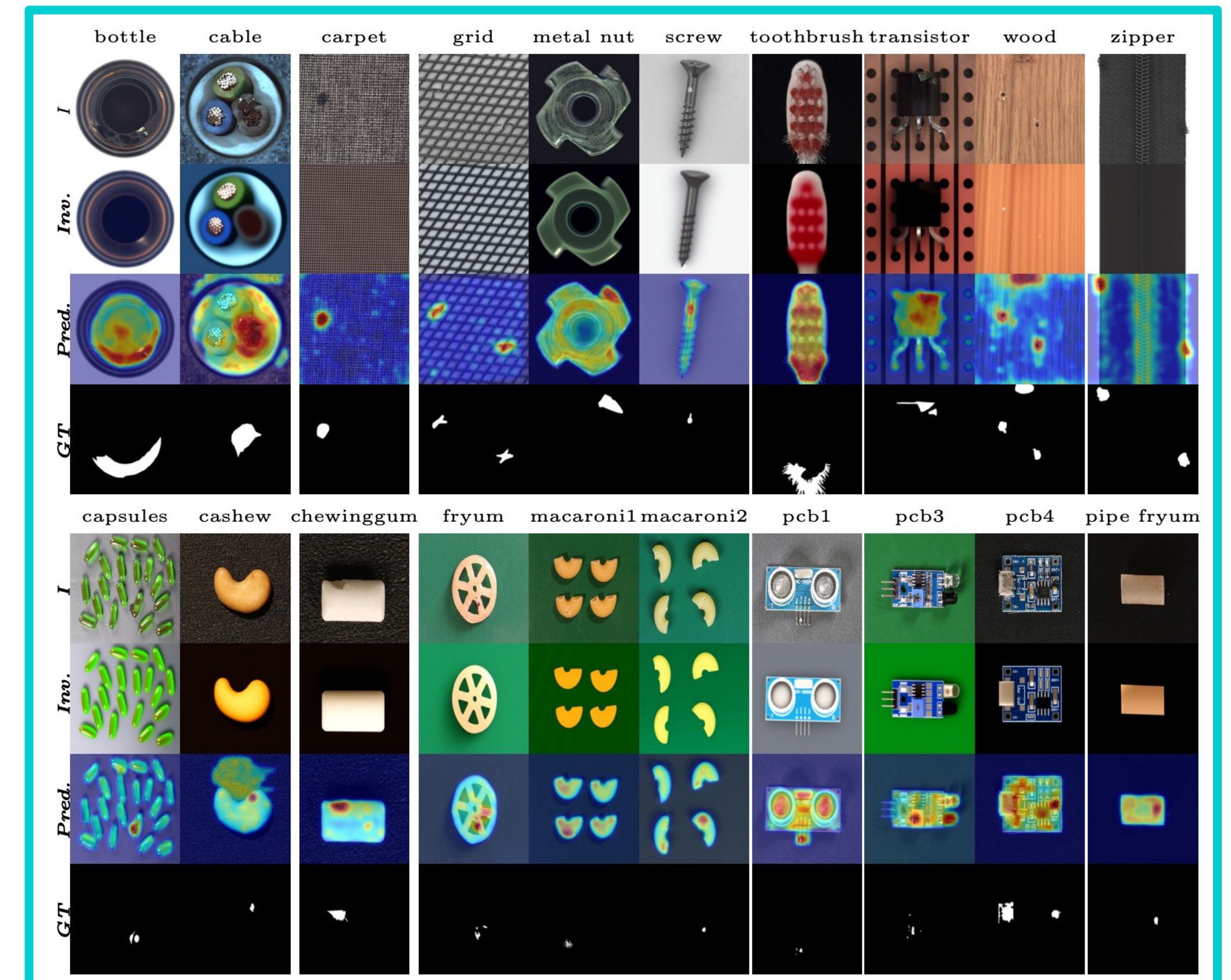
- Inversion** stage mirrors the forward process in reverse, from a clean latent code z which is obtained by encoding a given image, toward a noisy latent z_T .
- Denoising** takes inverted latent and feeds to the SD UNet model for denoising.
- Denoised latent** is given to image decoder to recover a clean normal image.
- Anomaly segmentation** stage takes input and generated images and performs segmentation on the feature space.

Results

Quantitative Results (MVTec-AD and VisA)

Method	Train	Guided Prompts	MVTec-AD					VISA				
			ROC_I	ROC_P	PRO	AP_P	$F1_P$	ROC_I	ROC_P	PRO	AP_P	$F1_P$
APRIL-GAN [11]	✓	✓	86.1	87.6	44.0	40.8	43.3	77.5	94.2	86.6	25.8	32.3
AnomalyCLIP [60]	✓	✓	91.5	91.1	81.4	34.5	39.1	80.9	95.5	86.7	21.3	28.3
FiLo [20]	✓	✓	91.4	93.1	61.7	52.2	52.0	83.9	95.9	85.4	31.4	38.5
PromptAD [35] †	✓	✓	90.8	92.1	72.8	-	36.2	N/A	N/A	N/A	N/A	N/A
AdaCLIP [9]	✓	✓	89.9	89.9	44.0	41.6	43.9	86.2	95.8	51.0	31.5	37.1
Bayes-PFL [42]	✓	✓	92.3	91.8	87.4	48.3	-	87.0	95.6	88.9	29.8	-
DZAD [58] †	✓	✗	93.5	86.7	-	-	-	90.2	92.0	-	-	-
WinCLIP [29]	✗	✓	91.8	85.1	64.6	18.2	31.6	78.1	79.6	56.8	5.4	14.8
ALFA [61] †	✗	✓	93.2	90.6	78.9	-	36.6	81.2	85.9	63.8	-	15.9
FADE [36]	✗	✓	90.0	89.6	84.5	34.1	39.8	78.1	91.5	79.3	10.0	16.7
AnomalyVLM [8]	✗	✓	62.4	73.1	42.4	28.7	37.7	71.0	75.8	36.9	13.8	22.1
RAG [44] †	✗	✓	92.5	89.1	-	-	-	86.8	87.6	-	-	-
MAEDAY [48] †	✗	✗	74.5	69.4	-	-	-	N/A	N/A	N/A	N/A	N/A
DIVAD (ours)	✗	✗	76.0	88.6	73.3	31.6	35.7	69.7	93.4	78.2	19.5	24.0

Qualitative Results (MVTec-AD and VisA)



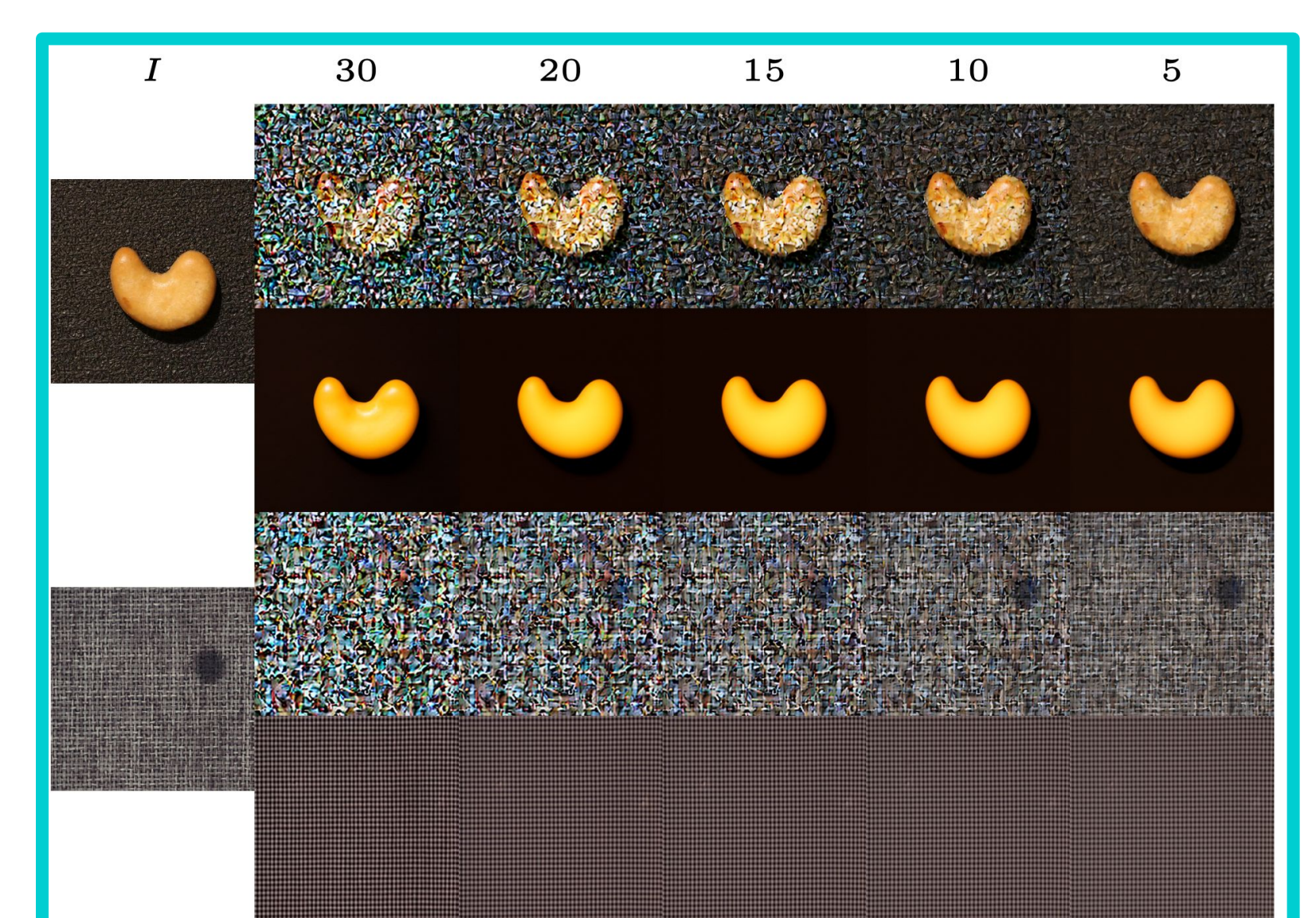
Effect of varying the timestep and SD version

Time step	SD vers.	MVTec-AD					VISA				
		ROC_I	ROC_P	PRO	AP_P	$F1_P$	ROC_I	ROC_P	PRO	AP_P	$F1_P$
30	1.5	70.9	87.7	71.1	28.0	33.6	64.0	92.4	74.3	15.6	19.2
20	1.5	72.2	87.9	72.5	30.0	34.8	69.1	93.4	77.8	18.4	23.4
15	1.5	74.2	87.9	72.5	30.7	35.2	69.4	93.4	78.1	19.5	24.1
10	1.5	74.4	87.8	72.2	30.8	35.2	70.2	93.4	78.3	19.7	24.4
5	1.5	73.9	87.2	70.9	30.5	35.4	71.1	93.3	78.2	19.7	24.6
30	2.1	72.5	87.9	71.1	28.1	33.1	65.8	92.5	74.6	14.9	18.6
20	2.1	74.4	88.0	72.8	31.0	35.4	68.2	93.5	78.0	18.3	23.0
15	2.1	75.1	88.0	72.6	31.0	35.4	68.7	93.5	78.2	19.5	23.9
10	2.1	76.0	88.6	73.3	31.6	35.7	69.7	93.4	78.2	19.5	24.0
5	2.1	75.3	87.5	71.4	31.0	35.9	70.1	93.3	78.2	19.2	23.9

Effect of object segmentation method

Dataset	Obj. Mask		Metrics				
	Obj.	Tex.	ROC_I	ROC_P	PRO	AP_P	$F1_P$
MVTec-AD	✗	✗	77.5	85.7	76.8	33.5	36.5
	✓	✗	76.0	88.6	73.3	31.6	35.7
	✓	✓	70.9	79.9	57.6	23.0	29.1
VISA	✗	N/A	67.5	85.7	67.6	13.0	17.9
	✓	N/A	69.7	93.4	78.2	19.5	24.0

Inversion Visualization



- [1] Bergmann, P. et al., "Mvtec ad—a comprehensive real-world dataset for unsupervised anomaly detection", in CVPR, 2019.
[2] Zou, Y. et al., "Spot-the-difference self-supervised pre-training for anomaly detection and segmentation", in ECCV, 2022.
[3] Song, J. et al., "Denoising diffusion implicit models", in ICLR 2021.

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